Export Map 2

Land Reforms

- Data
  - Landless Households — [Socio Economic Caste Census 2011] — 30%
  - Negative
    - %person own less than 2 hectare [NSSO] — 90%
    - agri cencus 2018 farmer data
      - Landless — 7
      - Marginal — 75
      - Small — 10
  - Positive — Small and marginal owned area
    - 1961 — 20%
    - 2013 — 53%
  - In 1991-92, the bottom 50% of rural households owned only 3% of the total land, while the top 10% owned 54% — NCF
  - Fragmentation: The average size of operational holdings has declined significantly, from 2.67 hectares in 1960-61 to 1.34 hectares in 1990-91 — NCF
  - In 1991-92, 11.24% of rural households were strictly landless — NCF
  - Currently, small and marginal farmers own 86% of the land in India — Model Contract Farming and Services Act, 2018
- Introduction
  - The agrarian economy of India has historically been characterized by inefficiency, low productivity, and small units of production
  - Agrarian Reforms Committee (Kumarappa Committee) recommended in 1949 that there should be no place for intermediaries and that "land must belong to the tiller". It viewed land reforms as the "touchstone of democracy" essential for freedom from poverty
- Issues — The colonial land tenure systems—Zamindari, Mahalwari, and Ryotwari—created a disconnect between the cultivator and the land, leading to exploitation
- Reforms so far
  - Abolition of Intermediaries (1950-1972): Removing Zamindars and Jagirdars to bring cultivators in direct contact with the State — Intermediaries were abolished over 6.0 million hectares, and ownership rights were granted to 20 million tillers
  - Tenancy Reforms: Providing security of tenure, regulating rents (recommended at 20-25% of gross produce), and conferring ownership rights to tenants
    - Rights of 12.42 million tenants were protected over an area of 6.25 million hectares
    - successful in kerala and WB
    - Operation Barga: In West Bengal Operation Barga was launched in 1978 with the objective of achieving the registration of sharecroppers and providing them permanent occupancy and heritable rights and a crop division of 1:3 between landowner and sharecropper.
  - Land Ceiling Acts: Fixing statutory limits on land ownership and redistributing surplus land to the landless — Redistribution: An area of 2.12 million hectares of ceiling-surplus land was distributed to 5.67 million rural poor (51% of whom were SC/ST)
  - Consolidation of Holdings: Merging scattered landholdings into compact blocks to improve farming efficiency — About 65.34 million hectares of land was consolidated, notably in Punjab, Haryana, and Uttar Pradesh
  - MoRD
  - Forest Rights Act, 2006: Recognized rights of forest-dwelling Scheduled Tribes and other traditional forest dwellers
  - Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013 (LARR Act): — Replaced the colonial 1894 Act to ensure humane, participatory processes and fair compensation for land acquisition
  - Recent steps
    - Model Agricultural Land Leasing Act, 2016: Proposed by NITI Aayog to legalize land leasing, ensuring security for land owners and access to credit for tenants
    - Model Land Titling Act: The central government has drafted a Model Land Titling Act, aiming to standardise procedures and facilitate a shift towards conclusive titling.
    - The Registration Bill 2025: Aimed at streamlining land transfers and ensuring that registration reflects actual ownership, not just transactions
    - DILRMP (Digital India Land Recored Modernisation Programme)
      - Digital India Land Records Modernization Programme (DILRMP): Aimed at digitizing land records
      - ULPIN / Bhu Aadhar — 14 digit alpha-numeric for each land parcel
      - GPS Based
      - integerated with land records
      - As of recent data, 94% of Rights of Record (RoR) have been computerized — 2024, MoRD
    - PM SVAMITAVA
      - 6.62 lakh villages in the country which will be eventually covered in this scheme.
      - clear ownership of property in 'Abadi' in habited area
      - It is a scheme for mapping the land parcels in rural inhabited areas using drone technology and Continuously Operating Reference Station (CORS) to provide property cards.This Scheme comes under the Ministry of Panchayati Raj.
- Factors Responsible
  - reforms succeeded (e.g., West Bengal's Operation Barga, Kerala),
    - Strong political commitment allied with peasant movements challenged the resistance of landed interests
    - Record of Rights: Successful states prioritized the updating of land records (e.g., recording sharecroppers in West Bengal)
    - Decentralization: Involvement of Panchayati Raj Institutions (PRIs) in identifying beneficiaries and managing land
    - Kisan Sabhas and Farmers Associations also helped farmers organize themselves and raise their demands.
    - Judicial backing and progressive interpretations of constitutional provisions aided in land reforms. Egland reform legislations through IX schedule.
  - Failure
    - Lack of Reliable Records: The absence of accurate land records made it difficult to identify surplus land or protected tenants
    - Legal Loopholes & Litigation: Large amounts of surplus land remained locked in court cases. Landowners used loopholes like Benami transactions to evade ceiling laws
    - Administrative Apathy: The revenue administration often lacked the vision and dynamism required for a holistic view of land management — NCF
    - Alienation of Tribal Land: There has been widespread dispossession of Dalits and Tribals from lands legally allotted to them
- Land Reforms and Agriculture Productivity
  - Debate
    - higher farm szie enhances productivity due to mechanisation, and economy of scale
    - small farm size increase productivity due to resource efficiency, high labor intensity and supervision
    - Productivity is linked to the size of holding; Indian productivity is much lower than global peers (e.g., paddy yield is 2929 kg/ha in India vs. 6321 kg/ha in China)
  - Reforms — Security of tenure encourages investment in land improvement. However, excessive fragmentation (minuscule holdings) hampers modern technology adoption To address this, contract farming and producer organizations (FPOs) are recommended to aggregate produce without dispossessing farmers of their land ownership
- Land Records
  - Issues
    - Records are often outdated and do not reflect ground realities (e.g., sharecroppers not recorded). — 60% of holdings still lack "conclusive titles," leading to endless litigation.
    - Discrepancy between revenue records and registration data — NCF
    - Slow progress in the survey/resurvey of land parcels
    - multiple claims — Complex inheritance and land transfer systems often result in overlapping claims and ownership ambiguity.
  - Land tilting
    - Current Status (Presumptive Titling)
      - India follows a system where land records are fiscal (for revenue) and presumptive.
      - Registration of deeds under the Registration Act (1908) records the transaction, not the title itself.
      - Ownership is proven through a chain of documents
    - Reforms Required (Conclusive Titling): There is a move toward Conclusive Titling, where the State guarantees the title. This requires a single, integrated agency for maintaining land records
      - Benefits
        - Reduced Litigation: More than 55% of civil disputes in India are land-related. Clear titles would drastically reduce this litigation
        - Access to Credit: Clear titles facilitate easier access to institutional credit (mortgages)
        - Greater Market Efficiency: Accurate and transparent land records can facilitate land transactions and promote a more efficient land market.
      - Challenges: The transition requires massive investment in surveying, updating records, and legal changes.
- Current Issues
  - Restrictive Tenancy Laws: Bans on leasing in many states prevent land from being used efficiently and leave tenants without legal protection — Kerala
  - Exclusion: Women, despite legal rights (Hindu Succession Amendment Act 2005), often lack actual control or titles (Joint Pattas) — women constitute only 14% of all operational landholders
  - Financial Exclusion: Tenant farmers cannot access formal credit (Kisan Credit Cards) due to lack of land titles
- way ahead
  - legal front
    - Legalize Land Leasing: States must adopt the Model Agricultural Land Leasing Act, 2016 to allow land owners to lease out land without fear of losing ownership and to enable tenants to access credit/insurance
    - Special Land Tribunals: Establish fast-track tribunals to resolve the 7.4 lakh acres currently locked in litigation (Economic Survey 2024-25 context).
    - Inclusive Definition of Farmer: Delink the definition of "farmer" from land ownership to include tenant farmers, sharecroppers, and women for accessing government benefits
  - Innovative
    - Land Pooling: Promote Farmer Producer Organizations (FPOs) and group farming to create economic scales of operation for smallholders
    - Contract Farming: Adopt the Model Contract Farming Act to link farmers with markets and technology while protecting land rights
  - Digitization: Complete digitization of land records on a "Mission Mode" including geo-tagging to reduce disputes
    - Digitizing Maps: Completing the 68% digitization of cadastral maps (DoLR 2024) to accurately identify encroached or surplus government land.
    - Administrative changes at the state level that streamline the collection and maintenance of land data, for example - Karnataka's BHOOMI Project.
  - Draft National Land Reforms Policy: The draft has five goals
    - Restore land unjustly taken from vulnerable communities such as Dalits (untouchables) and Tribals.
    - Protect the land of the Dalits and Tribals including the commons that they depend on going forward
    - Liberalize leasing laws.
    - Improve land rights of women.
    - Appropriate amendment to Forests Right Acts, 2006.
- Implementation — Good Implementation
  - Kerala (EMS Nambodiriprasha)
  - WB (CPIM)
- Challenges in digitisation

Main Topic 18

Horti/livestock/aquaculture/food processing

- % of GVA
  - 33
  - 30
    - 2014-15 — 24%
    - 2022-23 — 30%
  - 6.7
  - FPI
    - agri — 11.5%
    - manufacturing — 10%
- CAGR (Decadal)
  - 2.82% (Production wise)
  - 7.13%
  - 6.75%
  - 5.35% (2015-23)
- % of Agri exports
  - 7.5
  - 9.3
  - 15
  - 23%
- Employement
  - -
  - 21 Mn
  - 28 Mn
  - 70L / 12% of formal workforce
- Mission
  - Horti
    - Mission for Integrated Development of Horticulture (MIDH) — fruits,vegetables, flowers from prod to post harvest
    - National Horticulture Board (NHB) — post harvest infra in integrated manner
    - horticulture CDP
  - Livestock
    - NDDB — promote farmer owned dairy and cooperatives
    - Rashtriya Gokul Mission — indigenous bovines and enhance milk produciton and producivity
    - National Livestock Mission — sustainable dev of livestocks
  - Aquaculture
    - PM Matsya Sampada Yojna — Blue revolution
      - production, quality
      - sustainable and responsible
    - Fisheries and Aquaculture Infrastructure Development Fund (FIDF) — fish inra — harbours to cold chains
- Concerns
  - Horti — Climate change, water scarcity, post-harvest losses
  - Livestock — Greenhouse gas emissions, resource depletion, low productivity
  - Aquacultre — Environmental impact (mangrove loss), water pollution, disease outbreaks

Main Topic 19

Food Processing

- Intro — We export what we should eat and waste what we shoudl Process
- Conclusion — Backbone of AtmaNirbhar krishi, potential to linking farms into international food chains and reduce rural distress.
- GDP Contri
  - Agriculture — 11.5%
  - Manufacturing — 10%
- CAGR (2015-16 - 2022-23)
  - 5.35%
  - 1.61 -> 1.92 L Cr.
  - MOSPI
- % of total agri exports
  - 23%
  - Rising CAGR @13%
  - 13 Bn in 2022-23
- % of total organised employement
  - 12%
  - 70L
- Investment growth in last 7 year — 7.25%
- Budget — National Institute of Food Technology, Entrepreneurship and Management to be set up in Bihar
- Post Harvest Losses in Perishables — 20%
- Issues
  - Low Level of Processing
    - India — 10% of agri produce
    - Developed countries — 60-80%
  - Fragmented supply chains — lack of storage — 37 mt
  - lack of processing infra — testing labs, effluent treatment plants, modern processing units, etc.
  - Credit access — 6% of MSME in food processing recive it
  - Lack of Skilled Manpower — 80% unskilled/ informal — NSDC , 2023
  - Red Tapism
- Govt Stance
  - Economic
    - FDI — 100% thtough automatic route
    - PLI
      - in food processing sector (Mozarella Cheese)
      - In
        - Ready to eat
        - Millet
    - PSL
    - Low GST — 0/5%
  - Operation Green
    - Originally for — TOP
    - Now expanded to — (TOTAL) 22 perishable items
    - Aim — Price Stabilisation
    - method — Integerated logistics development
      - support of FPO
      - modernise agri logistics
      - processing facilities
      - managagement
  - PM-Kisan sampada Yojna
    - Cold storage, food parks, etc.
    - Investment — 10,000 Cr
    - Jobs generated — 6L
  - Market
    - ODOP
    - e-NAM
    - Agri Infra Fund
      - 1L Crore
      - Farm gate Infra
      - Primary Processing
- Need to be Done
  - Agri-export zone
  - Cluster Developement — JackFruit Cluster Kerala
  - Credit acess
  - Strengthen FPO and contract farming
  - Skill Development under Skill India Mission
    - Food Safety
    - Packaging

Farmer Debt

- % of household spend more than they earn — 70%
- % in debt due to rise in farm input cost — 50%
- NSSO data

Irrigation

- land under irrigation of Gross Cropped Area
  - 2016 — 49%
  - 2021 — 55%
  - NITI Aayog
- rice and sugarcane take — 60% of water available for irrigation
- Rainfed Agriculture — % of Prduction
  - Millets — 90%
  - Oil Seeds — 80%
  - Livestock/Cotton — 60%
- %of irrigated area under groundwater irrigation — 60%
- %of ground water usage for irrigation — 90%
- % of global fresh water resource in India — 4%
- Solution
  - Microirrigation can save upto [Ashok Dalwai Committee] — 40% of water
  - Major cause of water inefficiency [NITI Aayog]
    - Electricity subsidy
    - Use graded Metered electricity
- PMKSY
  - Accelerated irrigation benefit Programme
  - per drop mroe crop
  - Har Khet ko Pani — command area developemnt
  - integerated watershed development

Cropping Intensity

- 1950 — 111%
- 2019-20 — 151%
- Area Under Food Grain (Cross Cropped Area)
  - 1980 — 73%
  - 2022 — 62%
- Coarse cereals Area
  - 1980 — 22%
  - 2022 — 9%
- Oil Seed
  - 1980 — 9%
  - 2022 — 15%
- HortiCulture
  - 2001 — 8.5%
  - 2022 — 14%
- MoAFW

Crops and produce

- Cereal
  - 2023 — 332 mt
  - 1947 — 50 mT
  - Expected rise this year
    - 6.6%
    - 354 mt
  - Productivity
    - Wheat — 3.4-3.6 Ton
    - Rice — 2.8 Ton
    - India — ~3 tonne / hectare
    - China — ~6.1 tonne/ hectare
    - FAO
- Millets
  - Demand per capita per month in rural india
    - 1962 — 4.4Kg
    - 1983 — 3.3 Kg
    - 2011 — 0.4Kg
    - 90% decline from pre green reovlution
  - Export — 2nd Largest in world
  - Productivity — 1.9 tonne per hectare — MoAFW
  - Rise as IYoM capagain — 40% growth in sales — Amazon
  - why millets
    - Climate Resilence — <30 cm rain
    - staple crop for tribals — 60% — SECC
    - Benefits — Health
      - Gluten free ;
      - high nutrients
        - Ragi — 10x calcium than wheat or Rice
        - Bajra — Iron
      - low glycemic index
      - Anti-oxidants
  - Conclusion — Poor Man Grain -> Smart Food
- Horticulture
  - 352 mt — 2023
  - 2nd largest in world Production
  - pick 4 vital stats from above + change in area
  - Why a good sector
    - High income — 5-10x in apple relative to wheat
    - Labour Intensive — planting, harvesting, etc.
    - Nutritional Security — Hidden Hunger
    - Exports — $3.8 Bn
  - Challenges
    - Low Processing
      - 2-3% in fruits — MoSPI
      - Global — 60%
    - Post Harvest Loses — 20%
    - Fragmented Supply Chain
    - Price Volatility
    - Credit Flow — <5% of total agri credit
    - Climate Change Vulnearabilty
  - Govt Stance
    - MIDH (subsumes NHM)
      - production
      - infra
      - market
    - National Horticulture Board
      - Financial assistance for cold storage
      - Nurseris
    - Operation Greens : TOP -> TOTAL (incudes fruits and vegetables) — subsidies on transport and storage
    - Horticulture CDP
  - Way Forward
    - CDP — Kerala CDP Jackfruit
    - FPO promotion and contract farming — Banana in Jalgaon
    - Digitisation
      - Pest forecasting
      - AI Advisory
      - Kisan Sarthi App
    - Export oriented porduction — Grapes/Litchis
    - Cold Storage — through solar power
    - Crop Insurance
- Pulses
  - Protein by weight — 20-25%
  - Largest
    - Producer — 25%
    - Consumer — 27%
    - Importer — 14%
    - Import dependence — 25%
  - Solution: — Integrated Schemes of Oilseeds, Pulses, Oil palm and Maize or ISOPOM,
- Oil seeds — Import Dependence — 55%

Parlance

- water intensive — water guzzling crop
- dropiing water levels — conversion of indo-gangetic plain into dark zones
- 'bait to plate' tracability using IT technology in aquaculture
- human back to machine back via shift from bag packing to containerisation
- fork to farm
- Evergreen revolution coined by MS Swaminathan
  - to address shortcoming of Green Revolution
  - producing more with less land, less water, less pesticides
  - to achieve sustainiable agriculture
- NCF National commssion on Farmers — swaminathan
- DFI — dalwai committee
- high level committe — HLC — shanta kumar

Presented with xmind